

Pablo Leyva

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EDUCATION

New Jersey Institute of Technology

Newark, NJ

B.S. in Applied Mathematics (Statistics) & Computer Science (AI: Learning and Reasoning)

Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Software Engineering, Database Systems, Intro to Machine Learning, Neurosymbolic AI, Explainable AI, Deep Learning, Reinforcement Learning, Probability, Linear Algebra, Convex Optimization

EXPERIENCE

Apple

May 2026 – August 2026

AI/ML Intern

Cupertino, CA

- Worked on Agentic Commerce Siri: built full-stack **commerce agent harness** utilizing Apple Pay and Wallet APIs to execute multi-step agentic workflows for purchaing, booking, reserving, and scheduling and delegated sub-agents for "offline" work
- Designed **RL environments with verifiable rewards** (dynamic agentic commerce flows); implemented **GRPO and PPO** on-policy training with vLLM rollouts and a reward model trained on user preference data

Undergrad Research and Innovation (URI) - Agentic Reasoning SLM

Sep 2025 – May 2026

Research Fellow

Newark, NJ

- Trained a **3B-parameter reasoning SLM** via **SFT** → **DPO** → **RL fine-tuning** with multi-GPU distributed training (FSDP) on math-proof and logic-puzzle datasets, studying one-step reasoning generalization in small language models
- Constructed a **synthetic RL environment** with programmatically-checkable rewards for mathematical proof search; ran **ablation studies** across reward-shaping strategies and data-mixing ratios to isolate sources of reasoning gain
- Ran **linear probes** on hidden activations and **safety evaluations** (refusal consistency, reward hacking, out-of-distribution robustness), benchmarking the SLM against larger LLM baselines on lm-eval-harness

Apple

May 2025 – August 2025

AI/ML Intern

Cupertino, CA

- Built an **agent orchestration system** integrating the Apple Foundation Model (AFM) with Model Context Protocol (MCP); designed stateful multi-step tool execution with retry and human-in-the-loop fallback paths, collecting **trajectory-level preference data** across 10k+ rollouts
- Built the team's **LLM evaluation pipeline** benchmarking agent trajectories (tool-use success, groundedness, hallucination rate) against Claude and GPT baselines; results directly informed fine-tuning priorities
- Designed **tool selection and execution** flows with LangGraph using TouchID human-in-the-loop validation; presented the prototype at the team's end-of-internship review (sr dir offered return internship)

Caterpillar Inc.

May 2024 – August 2024

Software Engineering Intern

Chicago, IL

- Built an **end-to-end data pipeline** with Python and Apache Airflow (scheduled DAGs with retry and failure handling) ingesting 500K+ daily records; served model predictions via **FastAPI microservices** at sub-100ms latency for 10K+ daily requests
- Fine-tuned **transformer LLMs** (BERT, GPT-3.5) with **LoRA** on a 500K-record corpus; ran ablation studies on data mixing and learning-rate schedules, reaching 89% classification accuracy

PROJECTS

Mechanistic Interpretability of Reward Models | *PyTorch, TransformerLens, sae-lens, Skywork-Reward, hh-rlhf* | **Python**

- Recovered ~8 pts of preference accuracy on held-out adversarial pairs by training **sparse autoencoders** on reward-model activations, ranking features by causal influence on reward score, and ablating at inference the **reward-hacking proxies**

NeuroCache: KV-Cache Memory Manager | *PyTorch, ONNX, llama.cpp, WebAssembly, Transformers.js* | **C++**

- Implemented a **KV-cache memory manager** with attention-scored **eviction policy**, slot-based allocation, and cache-replacement algorithms for dynamic context compression; benchmarking memory efficiency and response quality across long conversation sessions

Self-Aligning Agent Harness | *PyTorch, TRL, Gymnasium, vLLM, LoRA, Weights & Biases* | **Python**

- Built an **agent harness that aligns its own policy**: self-critique against a natural-language constitution **self-generated preference pairs (RLAIF)** iterative **DPO** on a LoRA adapter over vLLM rollouts filtered by programmatic verifiers; ablated judge strength and preference volume while tracking reward hacking and refusal-consistency drift across alignment rounds

LEADERSHIP

President, Vice President - External Affairs, Event Coordinator

October 2023 – May 2026

Society of Hispanic Professional Engineers (SHPE)

Newark, NJ

- Built and shipped the chapter's first **React-Native app** in 27 days with a team of 5 engineers, growing it to **200+ users** in its first month

TECHNICAL SKILLS

Languages : Python, C++, JavaScript/TypeScript, Java, Swift

Systems & Infrastructure : Docker, Kubernetes, Git, PostgreSQL, MySQL, distributed training (DeepSpeed, FSDP), high-throughput inference (vLLM, Ray), REST FastAPI microservices, Apache Airflow pipelines,

ML Frameworks : PyTorch, scikit-learn, TensorFlow, Hugging Face Transformers, JAX, NumPy, pandas

Reinforcement Learning : PPO, GRPO, DPO, RLHF, reward modeling, Gymnasium, RL environment design, preference data

LLM Training & Evaluation : SFT, LoRA/QLoRA, fine-tuning, distillation, synthetic data, lm-eval-harness, safety evaluations, ablation studies